

Supply Chain Logistics Problem

Data Dictionary & Schema Reference

Source: Brunel University London / Kaggle

Dataset: [anisseezzebdi/supply-chain-logistics-problem](#)

1. Dataset Overview

This dataset was created by Brunel University London and is designed for supply chain route optimization problems, specifically to build Linear Programming (LP) models that minimize freight and warehouse costs. It models a real-world logistics network where customer orders must be routed from warehouses (plants) through ports to final destinations via freight carriers.

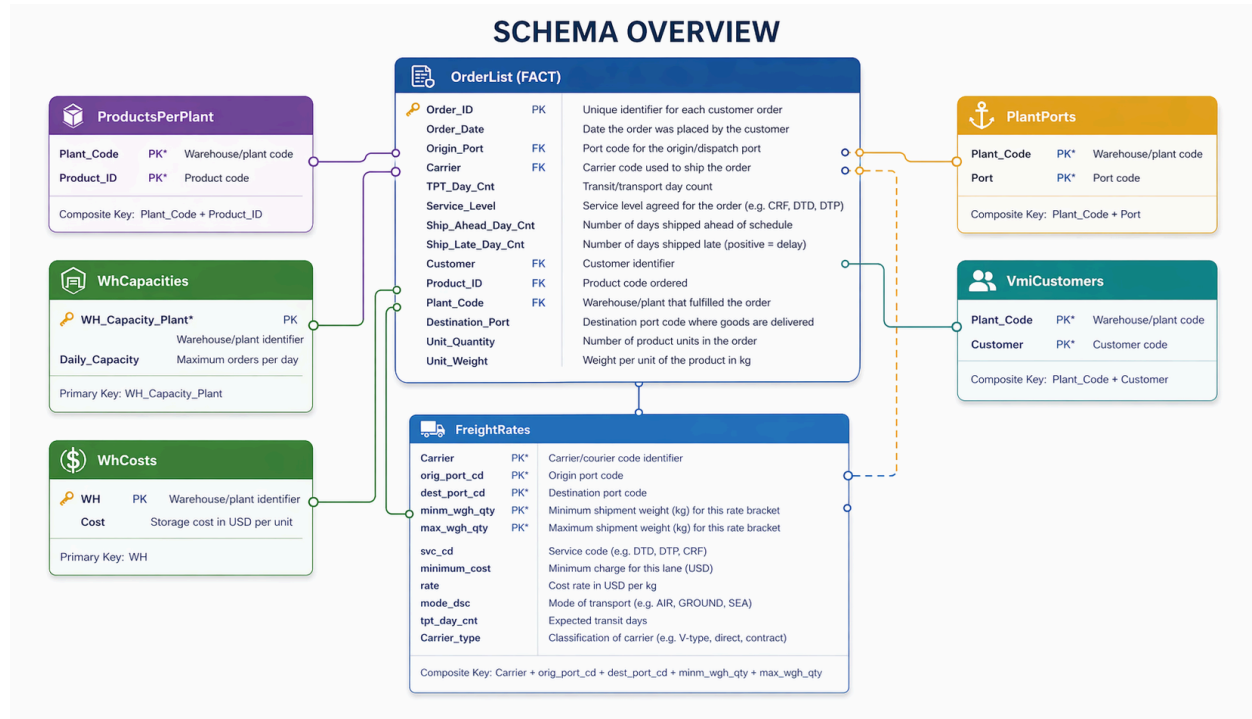
The dataset is structured around one transactional table (OrderList) and six constraint/reference tables that define the network's physical and commercial rules. Together, they allow analysts to compute the historical cost of routing decisions and optimize future routing under current constraints.

1.1 Dataset Summary

Table Name	Purpose	# Fields	Primary Key	Row Type
OrderList	Historical order records to be routed	~12	Order_ID	Transactional
FreightRates	Available couriers, lanes, weight brackets & rates	7	Carrier + Lane + Weight Range	Reference
PlantPorts	Permitted warehouse-to-port shipping links	2	Plant_Code + Port	Reference
ProductsPerPlant	Products that each warehouse is authorized to ship	2	Plant_Code + Product_ID	Reference
VmiCustomers	Warehouses restricted to specific VMI customers	2	Plant_Code + Customer	Reference
WhCapacities	Daily order capacity per warehouse	2	WH_Capacity_Plant	Constraint
WhCosts	Storage cost per unit per warehouse	2	WH	Constraint

2. Entity-Relationship Schema

The diagram below describes how the seven tables relate to one another. OrderList is the central fact table. All other tables act as constraint or reference dimensions that validate and cost-out the routing decisions recorded in OrderList.



2.1 Relationship Summary

The following table enumerates all foreign-key style relationships across the seven tables. Note that this is a denormalized analytical dataset, not a relational database. Relationships are enforced by business logic rather than database constraints.

Table	Foreign Key Field	References Table	Relationship Description
OrderList	Product_ID	ProductsPerPlant	Each order's product must be supported at the fulfilling warehouse
OrderList	Origin_Port	PlantPorts	Origin port on an order must be a permitted plant-port link
OrderList	Carrier	FreightRates	Carrier on order must exist in freight rates (some historical carriers may be absent)
OrderList	Plant_Code	WhCapacities	Plant/warehouse on order is subject to daily order capacity limits
OrderList	Plant_Code	WhCosts	Each plant has an associated storage cost per unit

Table	Foreign Key Field	References Table	Relationship Description
OrderList	Customer	VmiCustomers	Some plants are restricted to serving only specific VMI customers
ProductsPerPlant	Plant_Code	PlantPorts	Warehouses that stock products must also have valid port links
PlantPorts	Plant_Code	WhCapacities	All plants in port mappings have defined capacity constraints
PlantPorts	Plant_Code	WhCosts	All plants in port mappings have associated storage costs

3. Table Data Dictionaries

3.1 OrderList

Purpose

The central fact table. Contains one row per customer order with historical routing information — the plant (warehouse) used, the carrier, the port of origin, and shipping details. This is the primary input for LP optimisation: which product must be shipped, to which destination, in what quantity, at what weight.

Primary Key

Order_ID (unique identifier per order)

Fields

Field Name	Data Type	Key	Nullable	Description
Order_ID	VARCHAR	PK	No	Unique identifier for each customer order
Order_Date	DATE		Yes	Date the order was placed by the customer
Origin_Port	VARCHAR	FK	No	Port code for the origin/dispatch port. Links to PlantPorts.Port
Carrier	VARCHAR	FK	No	Carrier code used to ship the order. Links to FreightRates.Carrier (some historical carriers may be missing)
TPT_Day_Cnt	INTEGER		Yes	Transit/transport day count — the number of days in transit for this shipment
Service_Level	VARCHAR		No	Service level agreed for the order (e.g. CRF, DTD, DTP). Indicates speed/priority class
Ship_Ahead_Day_Cnt	INTEGER		Yes	Number of days the order was shipped ahead of schedule (buffer days)
Ship_Late_Day_Cnt	INTEGER		Yes	Number of days the order shipped late (positive = delay)
Customer	VARCHAR	FK	No	Customer identifier. Links to VmiCustomers.Customer for VMI restriction checks
Product_ID	VARCHAR	FK	No	Product code ordered. Links to ProductsPerPlant.Product_ID
Plant_Code	VARCHAR	FK	No	Warehouse/plant that fulfilled the order. Links to WhCapacities, WhCosts, PlantPorts, ProductsPerPlant
Destination_Port	VARCHAR		No	Destination port code where goods are delivered
Unit_Quantity	INTEGER		No	Number of product units in the order (used for cost calculation)
Unit_Weight	FLOAT		No	Weight per unit of the product in kg (used for freight weight bracket matching)

3.2 FreightRates

Purpose

Defines all available freight carriers, the shipping lanes they cover (origin port to destination port), the weight brackets they price, and the rates charged. Used in LP models to compute the cost of assigning a carrier to a given order on a given lane.

Composite Key

Carrier + orig_port_cd + dest_port_cd + minm_wgh_qty + max_wgh_qty (together uniquely identify a rate row)

Fields

Field Name	Data Type	Key	Nullable	Description
Carrier	VARCHAR	PK*	No	Carrier/courier code identifier (e.g. V44, V55). Links to OrderList.Carrier
orig_port_cd	VARCHAR	PK*	No	Origin port code for this freight lane. Links to PlantPorts.Port
dest_port_cd	VARCHAR	PK*	No	Destination port code for this freight lane
minm_wgh_qty	FLOAT	PK*	No	Minimum shipment weight (kg) for this rate bracket to apply
max_wgh_qty	FLOAT	PK*	No	Maximum shipment weight (kg) for this rate bracket
svc_cd	VARCHAR		Yes	Service code indicating transport mode or speed class (e.g. DTD, DTP, CRF)
minimum_cost	FLOAT		No	Minimum charge for this lane regardless of weight, in USD
rate	FLOAT		No	Cost rate in USD per kg for shipments within the weight bracket
mode_dsc	VARCHAR		Yes	Mode of transport description (e.g. AIR, GROUND, SEA)
tpt_day_cnt	INTEGER		Yes	Expected transit days for this carrier on this lane
Carrier_type	VARCHAR		Yes	Classification of carrier (e.g. V-type, direct, contract)

3.3 PlantPorts

Purpose

Enumerates the permitted connections between warehouses (plants) and shipping ports. A plant can only dispatch goods through a port that appears in this table. This is a hard network constraint used in route optimisation — any proposed route must use an allowed plant-port combination.

Composite Key

Plant_Code + Port

Fields

Field Name	Data Type	Key	Nullable	Description
Plant_Code	VARCHAR	PK*	No	Warehouse/plant identifier. Links to OrderList.Plant_Code, WhCapacities.WH_Capacity_Plant, WhCosts.WH
Port	VARCHAR	PK*	No	Port code that the plant is authorised to use for outbound shipments. Links to FreightRates.orig_port_cd and OrderList.Origin_Port

3.4 ProductsPerPlant

Purpose

Lists every valid warehouse-product pairing. A warehouse can only ship a product if that combination appears here. This prevents routing orders to plants that do not stock or handle the requested product.

Composite Key

Plant_Code + Product_ID

Fields

Field Name	Data Type	Key	Nullable	Description
Plant_Code	VARCHAR	PK*	No	Warehouse/plant code. Links to PlantPorts.Plant_Code, WhCapacities.WH_Capacity_Plant, WhCosts.WH
Product_ID	VARCHAR	PK*	No	Product code the plant is authorised to handle. Links to OrderList.Product_ID

3.5 VmiCustomers

Purpose

Vendor-Managed Inventory (VMI) customer exceptions. Any plant listed here is restricted to serving only the specified customer — it cannot supply other customers. Plants NOT listed in this table are unrestricted and may serve any customer. This is applied as a conditional constraint in LP models.

Composite Key

Plant_Code + Customer

Fields

Field Name	Data Type	Key	Nullable	Description
Plant_Code	VARCHAR	PK*	No	Warehouse/plant code with a VMI restriction. Only this plant-customer pair is allowed
Customer	VARCHAR	PK*	No	Customer code that the plant is exclusively permitted to serve. Links to OrderList.Customer

3.6 WhCapacities

Purpose

Defines the maximum throughput of each warehouse measured in the total number of orders it can process per day. This is used as an upper-bound constraint in LP optimisation to prevent over-assigning orders to a single warehouse.

Primary Key

WH_Capacity_Plant

Fields

Field Name	Data Type	Key	Nullable	Description
WH_Capacity_Plant	VARCHAR	PK	No	Warehouse/plant identifier. Links to OrderList.Plant_Code, PlantPorts.Plant_Code
Daily_Capacity	INTEGER		No	Maximum number of orders the warehouse can fulfil per day (capacity constraint for LP model)

3.7 WhCosts

Purpose

Specifies the unit storage cost at each warehouse, denominated in US dollars per unit stored. This is combined with order unit quantities in the LP model to compute the warehousing component of total supply chain cost.

Primary Key

WH

Fields

Field Name	Data Type	Key	Nullable	Description
WH	VARCHAR	PK	No	Warehouse/plant identifier. Links to OrderList.Plant_Code, PlantPorts.Plant_Code, WhCapacities.WH_Capacity_Plant
Cost	FLOAT		No	Storage cost in USD per unit stored at this warehouse (used in objective function of LP model)

4. Key Concepts & Analytical Notes

4.1 LP Model Inputs

To build a Linear Programming optimization model using this dataset, the following inputs are drawn from each table:

Source Table	LP Model Contribution
OrderList	Decision variables: which plant and carrier to assign per order; constraints on destination port and unit weight
FreightRates	Freight cost coefficient: $\text{rate} \times \text{unit_weight} \times \text{unit_quantity}$ per carrier-lane combination
WhCosts	Warehousing cost coefficient: $\text{cost per unit} \times \text{unit_quantity}$ per plant
WhCapacities	Upper bound constraint: total orders assigned to a plant $\leq$ daily_capacity
PlantPorts	Feasibility constraint: selected plant must be linked to the required origin port
ProductsPerPlant	Feasibility constraint: selected plant must be authorised to handle the ordered product
VmiCustomers	Conditional constraint: VMI-restricted plants may only be assigned to their designated customer

4.2 Known Data Quality Notes

Carrier V44_3 appears in OrderList but is absent from FreightRates. This is a historically used carrier whose rate contract is no longer active. When optimising for current constraints, this carrier should be excluded from the feasible carrier set.

VmiCustomers is an exceptions table — the absence of a plant from this table means it is unrestricted, not that it has no VMI relationship. Analysts should treat a NULL join result as 'no restriction applies'.

Unit_Weight in OrderList represents weight per individual unit. Total shipment weight for a freight bracket lookup is calculated as $\text{Unit_Weight} \times \text{Unit_Quantity}$.

4.3 Objective Function Structure

The total cost to minimize in the LP model is:

$$\text{Total Cost} = \text{Freight Cost} + \text{Warehouse Cost}$$

$$\begin{aligned}\text{Freight Cost} &= \sum (\text{FreightRates.rate} \times \text{OrderList.Unit_Weight} \times \text{OrderList.Unit_Quantity}) \\ \text{Warehouse Cost} &= \sum (\text{WhCosts.Cost} \times \text{OrderList.Unit_Quantity})\end{aligned}$$

4.4 Service Level Codes

Code	Description
CRF	Customer Referred Freight — customer selects and manages the carrier
DTD	Door-to-Door — full end-to-end service from origin warehouse to customer premises
DTP	Door-to-Port — shipment responsibility ends at the destination port

5. Glossary

Term	Definition
Plant / Warehouse	A storage and fulfilment facility that holds inventory and dispatches orders. Identified by Plant_Code
Port	A shipping port (sea or air) through which freight is dispatched. Warehouses connect to ports via PlantPorts
Carrier	A freight carrier/courier company that transports goods on a lane. Rate information is in FreightRates
Lane	A specific origin-port to destination-port route. Each lane can have multiple carriers and weight brackets
Weight Bracket	A range [minm_wgh_qty, max_wgh_qty] within which a specific freight rate applies for a carrier-lane pair
VMI	Vendor-Managed Inventory — a model where the supplier manages inventory levels at the customer's location. Some plants serve only their designated VMI customer
TPT Day Count	Transit/Transport Day Count — the number of days goods spend in transit from warehouse to destination
LP Model	Linear Programming Model — a mathematical optimisation technique used to minimise total freight and warehousing costs subject to the constraints defined across all 7 tables
DXA	Document formatting unit; 1440 DXA = 1 inch (used internally in document layout, not dataset-related)

*Dataset source: Brunel University London —
https://brunel.figshare.com/articles/dataset/Supply_Chain_Logistics_Problem_Dataset/7558679
Kaggle mirror: <https://www.kaggle.com/datasets/anisseezzebdi/supply-chain-logistics-problem>*